ECE 20007: Electrical Engineering Fundamentals I Lab

Spring 2026
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This syllabus is subject to change

1 Course Overview

Welcome the introduction lab course for Purdue Electrical and Computer Engineering (ECE) Labs. Throughout this course students will learn the skills required to measure, model, simulate, and design circuits. The skills students learn in this lab will carry forward into more advanced ECE Labs.

To develop these skills students will complete experiments designed to direct them in measuring and calculating many different values important to the fields of Electrical and Computer Engineering. Values such as voltage, current, resistance, and many others.

1.1 Course Objectives

A student who successfully fulfills the course requirements will be able to:

- Competently select, and use electronic measurement instruments.
- Design, wire, and troubleshoot electronic circuits.
- Simulate electronic circuits.
- Design, build, and test resistor-inductor-capacitor, operational amplifier, etc. circuits to meet specifications.
- Demonstrate knowledge of laboratory safety precautions.
- Maintain laboratory notes and create professional experiment reporting.

1.2 Required Material

- ADALM/Analog Discovery 2 (AD2) - If needed will be loaned form ECE Shop.
- The laboratory kit available at https://shop.eurekaleap.com/purdue-premium-ece-lab-kit-west-lafayette-campus/.
- USB flash drive. 4Gb or larger. Needed for connecting to bench equipment.
- [Optional] Digital Multimeter.
  Any handheld multimeter will work, as long as it has TRUE RMS capabilities. A multimeter with these capabilities will likely run $25-$45 USD on your preferred online retailer or local home improvement store.
- [Optional] Personal Oscilloscope Probes.

The optional items can be bundled along with the basic kit at special price. To take advantage of these special bundles, links are available below the product description at https://shop.eurekaleap.com/purdue-premium-ece-lab-kit-west-lafayette-campus/.

1.3 ADALM/Analog Discovery 2

If you would like to work on the lab curriculum outside of the lab space, an ADALM/AD2 will be assigned to through the ECE shop (BHEE 136), and you will be expected to return the same at the end of the semester. If you do not return, your student account will be charged for the cost of the device.

1.4 Course Schedule

Note: We reserve the right to modify the experiment order and topic. Any change will be announced via Piazza and with sufficient time to complete the prelab and prepare for class.

Session	Experiment	Prelab	Write-Up	Exam / Project	Report
0	Intro and Breadboards				
1	Ohm's Law		✗		
2	Kirchhoff's Laws and Voltage Dividers	✗	✗		
3	Nodal Analysis and LTspice	✗	✗		
4	Lab practical			✗	
5	RC circuits and Timing	✗	✗		
6	AC Signals and The Oscilloscope	✗	✗		
7	Passive Filters	✗	✗		
8	Operational Amplifiers	✗	✗		
9	Mini Project: The 555 Timer			✗	✗
10	Mini project Week 2				
11	Diodes and Protection Circuits	✗	✗		
12	Final project: Basic audio equalizer				
13	Final project Week 2	✗		✗	✗
14	Final project Week 3				

Table 1: Course schedule by session number.

2 Course Policies

The section Graduate Instructor is responsible for assessing the quality of your work and progress in the course. Ultimately they are responsible for determining your semester grade. Any section specific queries should be directed to your section GTA first.

To receive a passing grade, you must *at least*:

- Attend **all** scheduled lab sections (unless excused by the instructor). Students must request an excused absence strictly from the section GTA **before** the scheduled sessions begin.
- Perform all experiments.

- Submit a complete report for each lab.
- Obtain an average of 60% or better between the formal lab reports.
- Work with your partner(s) during the lab period.

Failing to complete any of the above criteria are grounds for failing the course.

Food and drink other than water are prohibited at all times in the lab. Anyone with open food or drink in lab will be required to leave. Water must be in a resealable container, and placed on the floor.

2.1 Piazza

Due to the numerous sections of the lab, any overreaching announcements, lab modifications, and assignment adjustments will be announced through Piazza, an online forum where students can ask questions, and instructors can post announcements. It will be expected that students check the course Piazza page regularly to make sure they stay up-to-date on the course. Email notifications can also be added for when instructors make announcements. Students initially enrolled in the course will be automatically populated into the course Piazza page. If a student is added late, or was not automatically populated into the page for whatever reason, they can join the course Piazza page by going to https://piazza.com/purdue/spring2026/ece20007.

2.2 Assignments

Lab Notebooks

Laboratory notebooks should be a log of all your work performed in this course. This notebook can be either digital or hand-written. **If you start with either digital or hand-written it needs to be consistent throughout the experiment.** The notebook should be updated in real time during the experiment. You will use the notebook notes to complete your write-up outside of lab hours.

The lab notebook should:

- Give a brief context to where you are in the experiment.
- Capture detailed results from experiment.
- Give **quick/short** insights into why you think an outcome occurs.

The lab notebook is essential to success in this course. It should be kept organized and neat to improve your effectiveness as researchers.

Prelabs

For all labs except the first lab, there is a prelab assignment that goes along with the lab. The Prelab Assignment for each lab consists of the following:

- Reading the reading portion for that week's lab.
- Completing the Prelab questions and simulations in your lab notebook
- Completing the Prelab quiz on Variate (Linked through Brightspace).

Write-Ups

For a majority of the experiments in the course you will need to turn in a digital version of an experiment write-up. This involves:

- **Short** description of each task including the results of that task
- A conclusion of the task based on the results
- Answer all questions in the experiments text
- Has **only** electronically drawn schematics and typed equations

Your write-up should be broken up into the tasks of the experiment. For each task you should have:

- Objective: 1-2 sentences describing the goal of the task.
- Digital schematic of the task
- The results of that task
- Calculations: Any calculations needed to be completed for the task.
- Conclusions: Use the results to identify whether or not an experiment was successful.

Reports

There will be **two** experiments that require a well documented formal report to be turned in. These documents will include in paragraph form:

- An abstract
- An introduction and conclusion to the overall problem and solution.
- A theory section that explains how the system and its parts work.
- Contains a discussion of the measurement results and conclusions made based on the results.
- Has **only** electronically drawn schematics and typed equations
- Answers all question in the experiments text

All Write-ups and reports need to completed in a word processor. Hand-written or stylus-written write-ups will not be accepted, Any hand-written or stylus-written component of the writeups and reports will receive a score of 0 for that submission. This includes hand-drawn schematics and figures.

Gradescope

All lab write-ups and reports will be submitted on Gradescope, a grading platform embedded inside Brightspace. Part of a proper submission will include selecting the appropriate pages of each assignment for each grade category. If pages are not selected, a grade deduction of 10% will be applied to your assignment grade.

2.3 Due dates

Prelabs

Prelabs are due by the start of your lab section. Late submissions of the prelab will not be accepted since students need to complete the prelab to be successful in the lab.

Write-Ups and Reports

Write-ups and reports are due at 11:59pm from one week of the day of your lab, skipping the first week of the semester. There will be a 20% deduction for each 24 hours late, starting at 12:00AM the next day, including weekends.

Late submissions for the final project report will not be accepted.

Lab reports turned in after midnight after the due date will be considered late.

Submission Policy

It is the responsibility of the student to ensure that submissions into the grading platform are on time and successful. If a student is having issues submitting an assignment for whatever reason, they should reach out to their Graduate Instructor in a timely manner to explain their situation. Late submissions will be discussed on a case-by-case basis.

Regrades

Requests for regrades must be made within one week of the grade in question begin returned to the student, and 24 hrs. After grades have been returned. Request made after this time will not be considered.

2.4 Cheating

- All work submitted must be the students own.
- The student **may not** copy text or figures from the course manual.
- Use of material from past semesters will be considered cheating.

All cases of cheating will result in a grade of **zero** on the assignment. Offense(s) may result in failing the course. Cheating may also be reported to the Office of the Dean of Students.

2.5 Large Language Models/AI/Generative Models

As generative models continue to embed their uses into our regular technologies, it would be a disservice to fully deny the use of the tool in modern education. Instead, the discussion should revolve around using the tools properly and ethically. The below concepts should be followed when using generative models in this course:

Effective Usage:

- Students are encouraged to use generative models to help brainstorm and organize thoughts. Especially for formal reports, generative models allow for multiple formats to be accessible in a short time period.

- Students are NOT permitted to use generative models to analyze data. This is a task that is expected to fully fall to the student. This is different than helping visually present data, as generative models can quickly provide multiple different ways to visually present data. It is still up to the student to determine the best way to present the data.

Ethical Usage:

- Students must be transparent where and how they use generative models in this course.

- Students may generate up to 20% of a submission using generative models. This should be focused on formatting and thought collection. All data analysis must still be completed by the student.

- Students must acknowledge where they use generative models in their writing using a reference note discussing the usage. A URL must also be provided to link to the prompts and outputs of the generative model. For example, the FAQ for ChatGPT link sharing can be found at https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq.

- Any writing or other submissions not explicitly identified as generated will be assumed as original to the student. Submitting generated work without identifying it as such will be considered a violation of Purdue's academic integrity policy, specifically the violation of "submitting a document written by someone else but representing it as one's own." As in any situation where a student is using writing from another person without correctly referencing the original author, this would be considered a form of plagiarism.

Generative models are still a developing technology, and how they are implemented in educational settings is still a divided topic of discussion, and can lead to grey areas on judgement. If a student is unsure on how much of their document is artificially generated, or if they are in violation of a policy, the student should reach out to at least 48 hours before their submission due date to receive guidance.

2.6 Grade Breakdown

This course uses a weighted grade system. The weight of each section of the final grade is listed below.

Write-Ups:	50%
Prelabs:	15%
Lab Notebook:	10%
Practical & Mini Project Reports:	15%
Final Project Report:	10%

Please note that all categories in this course are capped. There are several extra credit opportunities available, and these are divided into specific categories. Each category has a maximum weightage, and it is important to understand that these weightages will not exceed the capped limits listed above.

The following grade cut-offs will be used in this course.

$100\% > x \geq 96\%$	A+	$80\% > x \geq 76\%$	C+
$96\% > x \geq 93\%$	A	$76\% > x \geq 73\%$	C
$93\% > x \geq 90\%$	A-	$73\% > x \geq 70\%$	C-
$90\% > x \geq 86\%$	B+	$70\% > x \geq 66\%$	D+
$86\% > x \geq 83\%$	B	$66\% > x \geq 63\%$	D
$83\% > x \geq 80\%$	B-	$63\% > x \geq 60\%$	D-
		$60\% > x$	F

3 Course Expectations

3.1 Preparing for Class

The experiments for ECE 20007 require the full lab session to complete. To prepare for lab you should come to lab having already read through the experiment and answered all of the prelab questions. **The prelab questions are designed to help you with some of the more challenging experiment tasks you will encounter.**

3.2 In Class

During lab you should perform each task of the experiment. Recording your work in your lab notebook as you go. Your lab notebook is proof that you performed the experiments.

If you do not finish the experiment in class, you must complete it before the beginning of the next week's experiment.

3.3 After Class

After performing the experiment and gathering results, you will need to analyze the results and draw conclusions. Analyzing data should be done in tools such as MATLAB, Python NumPy, or any other numerical computing applications to make plots and compare results.

Your analysis should always compare theoretical calculations with measured results and attempt to measure and describe the causes of error.

3.4 Individual vs. Group Work

Prelabs

Prelabs are **individual** work. Working with others to formalize a strategy is okay. All questions must be solved on your own.

Write-Ups and the Final Project

Write-ups and the final project are a combination of group and individual work. Work done in the lab to complete specific lab tasks is considered group work. The data and

results are also group work and can be shared **as long as everyone was present and actively completing the task.** The written and analysis sections of the write-ups and final report are **individual** work.

Mini- Projects

In ECE20007, there will be a series of mini projects (both mandatory and optional bonus) that will be completed to test your **individual** skills and abilities in electronics. These labs will be **individual**. You will still be in pairs with your lab partner at your bench, but you are expected to make the circuits and measurements yourself with your own lab kit. You can communicate with your lab partner, TAs, or other students, but you are expected to work out the math, do the experiments, and complete the report as an **individual** assignment.

3.5 Attendance

You are expected to be present in lab to complete the experiments. If you cannot attend a lab for whatever reason, please reach out **only** to your section Graduate Instructor at least 24 hours prior to the lab starting so a solution can be determined.

Attendance will be taken throughout labs. Your participation and presence in labs dictate the granting of due date extensions, and additional non-emergency absences later in the semester. Having unexcused absences will greatly hinder the probability of being granted extensions and non-emergency absences.

3.6 Open Lab / TA Office Hours

Office hours will be held in (location will communicated later) throughout the day, accessible via a keypad code, and in BHEE 166 outside of lab hours. You will have access to these rooms as long as they remain clean and the equipment is not damaged. TAs' availability for office hours, both graduate and undergraduate, will be maintained on the digital calendar that can be found in Brightspace. To setup an office hour meeting with course coordinator, please send an email preferably at least 24 hours in advance.

Online TA office hours may be available between 7AM and 5:30PM every school day which will be found on the same calendar as the office hours. Students will be expected to use an ADALM/AD2 in order to be successful in online office hours.

3.7 Safety

There are several dangers that can be encountered in an electronics laboratory. These include (but are not limited to):

- Shock
- Burns
- Mechanical injury
- Chemical exposure

Be aware of these potential dangers and make every effort to avoid them. Violation of these rules may result in access privileges being revoked. There are cameras in the lab and we will review them as needed.
Lab instructors hold the responsibility of ensuring the lab is in a safe working condition before the students use the area, and enforce proper safety etiquette in the lab. If students are intentionally unsafe in the lab, encouraging unsafe behavior, or are continuing unsafe behavior after being told to cease the behavior, the student will be removed from the lab, resulting in a failing grade for that week's lab. If multiple occurrences of this behavior continue, more severe consequences can be discussed. These additional consequences can include but are not limited to:

- Failing of the course
- Reports to student advisor
- Reports to campus and local law enforcement

Equipment Safety

The benchtop equipment that is used in this lab is very expensive, and can be damaged if not used properly. If a student is found intentionally misusing the equipment, the same consequences can be issued as stated above for unsafe actions in the lab in addition to putting a hold on a student account until payment is made to replace any damaged equipment.

It is recommended students use the provided desktop computer in the lab when using the benchtop equipment. Students use their own personal devices such as laptops at their own risk when using the benchtop equipment. For example, If the final project is not made correctly, there is a chance of surges flowing back to the computer through the audio port of the computer, which can fry the audio port, motherboard and power supply unit.

3.8 Questions or issues

If you are in doubt about any section of this syllabus or the grading of the course, ask your TA. If you are dissatisfied with any aspect of the content or teaching of this course please inform your TA or the course coordinator.

4 Purdue honor pledge

Purdue's Honor Pledge was developed by students to advance a supportive environment that promotes academic integrity and excellence. It is intended that this pledge inspires Boilermakers of all generations to stay "on track" to themselves and their University. We encourage you to make this pledge. The pledge is:

> As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.

You may find more information at https://www.purdue.edu/provost/teachinglearning/honor-pledge.html

5 CAPS

CAPS Information: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

6 Non discrimination

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the

exchange of ideas, and enriches campus life. Purdue's nondiscrimination policy can be found at: http://www.purdue.edu/purdue/ea_eou_statement.html.

7 Students with documented disabilities

The Disability Resource Center (DRC) is a resource for students and instructors. Students may present a "Letter of Accommodation" at any point in the semester. Should you have questions about accommodations, please contact the DRC at: 494-1247 or drc@purdue.edu. In many cases the DRC can partner with you to develop inclusive teaching strategies that benefit all students in your class

8 Academic integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

9 Campus emergencies

In the event of a major campus emergency, course requirements, deadlines, and grade weights are subject to change. In case of such an event, information will be provided via Piazza and email.Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis. A link to Purdue's Information on Emergency Preparation and Planning: https://www.purdue.edu/ehps/emergency-preparedness/

This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. It is encouraged for students to download and review the Emergency Preparedness for Classrooms document.